

Executive Summary

India's ambition to enter the global race in large language models (LLMs) comes at a time of unprecedented activity worldwide. Over the past three years, US tech giants (OpenAI's GPT series, Google/DeepMind's Gemini, Anthropic's Claude, Meta's Llama) and Chinese firms (Baidu's Ernie, Alibaba's Tongyi) have rapidly released ever-larger models – with Google and Anthropic each unveiling new “Gemini” and “Claude” models in 2024–25 ¹ ² . Open-source efforts (Meta's Llama, HuggingFace's Bloom, EleutherAI's GPT-NeoX) have also scaled up dramatically. For example, in early 2026 Meta released Llama 3 (8B and 70B parameters) claiming “state-of-the-art performance” at those sizes ³ (see chart below). India's activity is growing: the government has launched the IndiaAI Mission (₹10,372 cr budget) to support computing infrastructure (38,000+ GPUs now onboarded) and sovereign models ⁴ ⁵ . Key initiatives include *BharatGen* (June 2025), an open-source multilingual LLM consortium, and private startups like Sarvam AI (Bengaluru) and Krutrim AI (Bangalore) building Indic-focused models ⁶ ⁷ . However, substantial gaps remain versus leading teams: Indian datasets ($\approx 10^{11}$ tokens) are orders of magnitude smaller than the $10^{13}+$ tokens used by GPT-4/Gemini; India has far fewer HPC GPUs and specialized researchers; and LLM training costs (tens of millions of USD) greatly exceed typical Indian project budgets ⁸ ⁹ .

This report analyzes the current global and Indian landscape of LLMs, identifying compute, data, talent, funding, and policy gaps. It proposes a detailed technical roadmap (model architectures, open-source toolkits, safety/alignment practices) and data strategy (sources, cleaning, augmentation for 20+ Indian languages) for building competitive models. We outline compute planning (on-prem vs cloud GPU options, estimated FLOPs and costs), team roles and partnerships (industry, academia, open source), and funding models (venture, consortium, public-private) to sustain the effort. We also address policy and ethics (data compliance, bias mitigation, government standards). Finally, we give a phased 3–5 year implementation plan with milestones, risks, KPIs, and a mermaid timeline. Practical action items for the first 90 days and year for an individual engineer – from experiments on small models to building collaborations – are provided to kickstart India's entry into the global LLM arena.

Global LLM Landscape: Major Models and Players

Timeline of recent key releases:

- **OpenAI (US):** GPT-3 (175B, 2020) sparked the current era. ChatGPT (based on GPT-3.5) launched Nov 2022. GPT-4 followed in Mar 2023 ¹⁰ . OpenAI then released GPT-4.5 (Feb 2025) – dubbed “our largest and best model for chat yet” ¹⁰ – and GPT-5 (Aug 2025, 1.8T param) ⁸ .
- **Google/DeepMind (US):** Launched PaLM (2022) and Bard (2023). Gemini models debuted Dec 2023 (v1), with Gemini 2.0 (Flash/Pro/Flash-Lite) rolling out Dec 2024–Feb 2025 ¹ . Google's models emphasize multimodal understanding and extremely long context (100k+ tokens).
- **Anthropic (US):** Released Claude 1.0 (Jun 2023) and quickly iterated: Claude 3 (Mar 2024), 3.5 (mid-2024), 3.7 (early 2025) etc ² . Claude models introduced features like 200K-token context windows and “computer use” plugins (e.g. browsing) by late 2024.
- **Meta (US):** Open-sourced LLaMA 1 in Feb 2023 and LLaMA 2 (70B) in July 2023. In April 2024 Meta unveiled Llama 3 (8B and 70B), claiming these “8B and 70B ... models are a major leap over Llama 2” ³ and deliver state-of-art scores at that scale. This open approach accelerated ecosystem development. (Meta also released other open models like Llama 3 Mixtral.)

- **Others:** Nvidia's xAI (US) released Grok (Nov 2023) aligned to public sentiment. European firms like Mistral (France, 2023) and AlephAlpha (Germany) trained specialized LMs. Cohere (Canada) and AI21 (Israel) are commercial LLM startups.
- **China:** Major tech firms have ChatGPT-style bots: Baidu's *ERNIE Bot* (closed beta Mar '23, public Aug '23) built on its ERNIE foundation models ¹¹, and Alibaba's *Tongyi Qianwen* (Apr '23) ¹². These support Chinese and major world languages. Research labs (e.g. SenseTime, Huawei) have or are developing their own models.

Meta Llama 3 Instruct model performance

	Meta Llama 3 8B	Gemma 7B - It Measured	Mistral 7B Instruct Measured		Meta Llama 3 70B	Gemini Pro 1.5 Published	Claude 3 Sonnet Published
MMLU 5-shot	68.4	53.3	58.4	MMLU 5-shot	82.0	81.9	79.0
GPQA 0-shot	34.2	21.4	26.3	GPQA 0-shot	39.5	41.5 CoT	38.5 CoT
HumanEval 0-shot	62.2	30.5	36.6	HumanEval 0-shot	81.7	71.9	73.0
GSM-8K 8-shot, CoT	79.6	30.6	39.9	GSM-8K 8-shot, CoT	93.0	91.7 11-shot	92.3 0-shot
MATH 4-shot, CoT	30.0	12.2	11.0	MATH 4-shot, CoT	50.4	58.5 Minerva prompt	40.5

Figure: Benchmark performance of recent LLMs. Meta's Llama 3 (8B, 70B parameters) outperforms previous open models of similar size ³ and rivals larger models on key tasks. This chart (from Meta) illustrates how competitive open-source models have become compared to proprietary peers.

The global pace is accelerating – almost every quarter sees new model releases with more parameters, data and compute. For example, epoch.ai estimates GPT-4 training used $\sim 2.1 \times 10^{25}$ FLOPs ¹³ (hundreds of GPU-years), while even smaller-scale models can cost millions: Llama 2 (70B) required 2048 A100 GPUs for 23 days ($\sim \$1.7M$) ¹⁴. In short, major companies invest tens of millions to billions of dollars and massive data to train these models, setting a high bar for any newcomer.

India's Current Efforts and Projects

India has launched several initiatives to catch up:

- **IndiaAI Mission (MeitY, 2024):** A national program with ₹10,371.92 crore approved (March 2024) ⁴ to build AI infrastructure and talent. Under *Compute Pillar*, IndiaAI has provisioned $\sim 38,000$ GPUs at subsidized rates (₹65/hr) for startups and researchers ¹⁵. Under *Foundation Models Pillar*, four startups were initially selected to build sovereign LLMs, including Sarvam AI (Bengaluru) ¹⁶.
- **Sarvam AI (Bengaluru):** A startup chosen by IndiaAI to develop India's first sovereign LLM ¹⁷. It has raised $\sim \$41M$ in funding ¹⁸. In Feb 2026 Sarvam unveiled two models: Sarvam-30B and Sarvam-105B (open-sourced under Apache 2.0) trained from scratch on Indic languages ¹⁹. The 105B model (Indus) has a 128k token context window and is aimed at enterprise use ²⁰. Sarvam reports using mixture-of-experts architectures and plans vision/speech modules too ²¹ ²².
- **Krutrim AI (Bangalore):** A startup creating "Krutrim LLM" focused on Indic languages. According to their paper, they aggregated "the largest known Indic dataset" (≈ 2 trillion tokens across languages) and trained a multi-expert model that matches state-of-the-art on Indian-

language benchmarks ⁷ . This shows private labs are pushing large-scale data collection for India.

- **BharatGen (IIT-Bombay consortium, DST, 2024–2025):** A government-funded, open-source initiative launched June 2, 2025 (BharatGen Summit). It aims to produce a homegrown multilingual generative model supporting 22 Indian languages ⁶ . BharatGen will train on “Bhartiya” datasets, emphasizing India’s cultural and linguistic diversity ⁶ ²³ , and share models and code publicly.
- **Academia and NGOs:** Research groups like AI4Bharat (IIT-Madras) have amassed large Indic corpora (251B tokens over 22 languages ²⁴) and built models (IndicBERT/BART, AIRAVATA LLM). The IndiaAI mission is funding *AIKosh*, a dataset platform with thousands of curated Indian data sources ²⁵ . Grassroots projects (e.g. Common Crawl’s IndicCC) and university labs (IIIT Hyderabad, etc.) also contribute open data and models for Indian languages.

These efforts signal momentum, but India’s ecosystem is still nascent compared to Silicon Valley or Beijing. In particular, corporate-tech involvement (beyond a few startups) remains low, and many projects are just now producing first results (e.g. Sarvam’s models). The coming years will reveal how well these initiatives scale.

Gaps vs. Global Leaders (Compute, Data, Talent, Funding, Policy, Ecosystem)

India must bridge significant gaps to compete with global leaders:

- **Compute and Infrastructure:** Leading LLM training runs on vast supercomputers or cloud clusters (tens of thousands of high-end GPUs/TPUs). For example, estimates put GPT-4’s training at $\sim 2 \times 10^{25}$ FLOPs (hundreds of A100 GPU-years) ¹³ . India’s new budget ($\sim 38k$ GPUs under IndiaAI) is substantial, but still far smaller in aggregate. Moreover, India lacks domestic GPU chip production; most hardware must be imported or leased. Existing Indian HPC (e.g. PARAM Shakti) are CPU-oriented and not optimized for neural networks. Without sustained investment, India risks being compute-constrained.
- **Data:** Global LLMs use colossal datasets (GPT-3 reportedly used $\sim 500B$ tokens; Google’s PaLM on $\sim 2T$ tokens). By contrast, open Indian-language corpora are tiny. For example, one analysis notes only $\sim 20B$ Indic tokens are in public crawls, versus 5–15 trillion tokens of mostly English web data ⁹ . Even combined Indian datasets (news, Wikipedia, etc.) are orders of magnitude smaller. This shortage makes pretraining from scratch challenging. India must aggregate content from government archives, publishers, web, and consider synthetic data (e.g. translate high-resource text, back-translate, or use existing LMs to generate data) to bulk up training corpora.
- **Talent:** India produces many IT grads and data scientists, but the specialized research talent (ML PhDs, AI ethics experts) is relatively scarce. Estimates suggest India’s entire AI workforce ($\sim 416,000$ in 2025) covers only ~ 50 – 55% of industry demand ²⁶ . The cutting-edge LLM work is concentrated in a handful of labs. Brain drain (talent moving abroad or to global giants) exacerbates the shortfall. Without rapid upskilling and incentives (e.g. IndiaAI fellowships of 500 PhDs by 2025 ²⁷), India may lack the experts to design and refine frontier models.
- **Funding:** Global tech companies spend freely – e.g. OpenAI’s Sam Altman has noted GPT-4 training cost “more than \$100M” ⁸ . By contrast, Indian budgets are an order of magnitude smaller. The entire IndiaAI Mission ($\text{₹}10,372$ crore) must support infrastructure, research, and startups over several years ¹⁷ . Startups like Sarvam have raised ($\sim \$40M$) ¹⁸ , but Indian VCs and tech firms are still building appetite for deep AI. Consortia or public-private models (like NVidia reportedly planning to invest in Sarvam ²⁸) will be crucial to finance expensive model training.

- **Policy & Regulation:** Global leaders have varying self-regulation (OpenAI's principles, EU AI Act drafting) and access to western data. India must quickly establish guardrails. The Bureau of Indian Standards is already drafting quality criteria for LLMs (e.g. requiring checks on caste/gender bias) ²⁹. India's upcoming Data Protection law (DPDP Bill) and copyright reforms will influence what data can be used. Ethical norms (fairness, explainability) must be codified early. The government also needs clear guidelines on privacy, security (prevent misuse), and export controls (e.g. HPC exports from US). Balancing open innovation versus national control will be a policy challenge.
- **Ecosystem:** In the West, vibrant open-source communities (HuggingFace, EleutherAI, BigScience) accelerate progress. India's ecosystem is smaller and more fragmented. There is a nascent community (e.g. hackathons, meetups like WACL India), but India needs more cross-disciplinary collaboration. Training developers in LLM tools, celebrating open contributions, and integrating efforts (e.g. AI4Bharat, GitHub sharing) will be important.

In summary, India has identified its weaknesses (compute scarcity, data paucity, talent crunch) and begun remedial programs. But globally, the LLM race has a steep learning curve. To catch up, India must combine its strengths (large developer base, linguistic diversity, innovative drive) with strategic investments and partnerships.

Technical Roadmap: Model Design and Open-Source Stack

An Indian LLM strategy should leverage modern best practices and open tools to minimize cost and maximize innovation:

- **Model Architecture & Scale:** State-of-the-art LLMs use decoder-only Transformers (GPT-like), often with enhancements. Options include dense decoder models (GPT, Llama), encoder-decoder (T5/PaLM), or Mixture-of-Experts (MoE) to economize FLOPs. For instance, Sarvam's models use MoE to have a sparse 9B "activated" parameters per token even in a 105B model. India should plan models at multiple scales: e.g. a 7–10B baseline model (trainable on moderate cloud resources) and a larger 50–100B variant (for later) depending on budgets. High-tokenization vocabulary (e.g. 128k tokens in Llama 3 ³⁰) and longer context (10k–100k tokens) are important goals. Architectural improvements (such as Rotary Positional Embeddings, GQA attention, FlashAttention) can boost efficiency.
- **Pretraining & Datasets:** The core recipe is unsupervised language modeling on massive text. Pretraining requires optimized data pipelines: deduplication, quality filtering (C4-like or "Pile" scripts), and tokenization (e.g. SentencePiece/BPE). Pretraining recipes often run for many epochs on huge text corpora. India can adopt open-source solutions: HuggingFace's `datasets` library for handling data; NVIDIA's Megatron-LM or Meta's Fairseq for large-scale training; Microsoft's DeepSpeed and ZeRO to shard large models across GPUs; and frameworks like PyTorch or JAX. Tools like Ray or Slurm can manage distributed training.
- **Instruction Tuning & Alignment:** After base pretraining, fine-tuning on instruction-response data yields models like ChatGPT. India should gather diverse instruction data (FAQs, educational dialogues, customer support transcripts in Indian languages) and use methods like supervised fine-tuning (SFT) followed by RLHF/Constitutional AI for alignment. Open-source RLHF toolkits (e.g. Hugging Face TRL, Apple's RL tools) can be used. Alignment safeguards are crucial: apply filters for offensive or sensitive content, and incorporate bias/difficulty prompts to expose issues. IndiaAI's focus on "Safe AI" and BIS standards ²⁹ suggests built-in bias testing. For example, multi-turn QA and chain-of-thought prompts should be evaluated in local contexts.

- **Evaluation:** Rigorous testing against benchmarks is key. Along with standard English tasks (MMLU, GSM8K, Hellaswag), India should use IndicGLUE/XTREME for Indian languages. Create new benchmarks for multilingual reasoning and domain tasks. Safety evaluations (e.g. the CivilComments bias dataset) should be extended to reflect caste, religion, and gender issues (as per India's needs ²⁹). Tools like eval harnesses (LM-eval-harness, VicunaEval) or custom test suites will be needed. Human evaluation (crowdsourced) on Indian prompts is also important.
- **Open-Source Stack & Tooling:** Use and contribute to open frameworks: Hugging Face Transformers, TensorFlow-Mesh for TPUs, PyTorch with NVIDIA Apex, etc. Adopt CI/CD and MLOps best practices: Docker containers (NVIDIA-docker), Kubernetes clusters, MLflow for tracking experiments, and monitoring (Prometheus). Data augmentation tools (imgaug for visual data, backtranslation for text) can enrich the corpus. Security tools (Adversarial filtering, logging) help detect misuse. Emphasize reproducibility: containerize training, document hyperparameters, publish code on GitHub/GitLab.
- **Modality Expansion:** After text models, consider vision and speech. Multi-modal models (like DALL-E or GPT-4 with vision) are the frontier. India's BharatGen is explicitly multimodal. Plan to integrate Indian-language speech data (e.g. from Bhashini or AI4Bharat's speech corpora) so models can input audio, text, and images. This requires adding encoders for images/speech (Vision Transformers, Whisper-based encoders) and joint pretraining. These are high-cost extensions, so they may come in later phases.

In summary, the technical path is to start with a robust open-source base: train or fine-tune models on Indian data using community frameworks, then layer on Indian-specific enhancements (tokenizer vocab with regional scripts, prompt libraries of Indic questions, alignment rules). Open-source is crucial: given resource constraints, building on community platforms (like HuggingFace Hub, collaborating with Eleuther/HuggingFace India) will accelerate progress.

Data Strategy for Indian LLMs

High-quality, diverse data is the fuel for LLMs. Key elements of an Indian data strategy include:

- **Data Sources:** Collate text from *public domain and permissively licensed* sources: Wikipedia (22+ Indian languages), government documents (laws, court judgments, parliamentary records), news archives, books (Project Gutenberg in Indian languages), and web crawl (Indic web). Coordinate with the *AIKosh* portal, which integrates thousands of Indian datasets ²⁵. Leverage Bhashini's corpora (translation pairs, speech transcripts) for multilingual alignment. Partner with publishers/telecoms for anonymized user queries or chat logs (with consent). Use APIs (e.g. OpenSubtitles, CommonCrawl's Indic extracts).
- **Licensing and Compliance:** Prioritize CC-BY/CC-BY-SA texts to avoid legal issues. Government data (railways manuals, census text, etc.) is often open. For copyrighted material (news, books), obtain licenses or use fair-use interpretations carefully. Maintain provenance metadata. Where needed, apply differential privacy or anonymization (remove personal info) to meet Indian laws (once DPDP is enacted) and global norms (GDPR).
- **Multilingual & Code-Mixing:** Ensure coverage of all major Indian languages (Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, etc.) and regional dialects. Include transliterations and code-mixed text (e.g. Hinglish chat conversations). Generate synthetic parallel data by back-translating between Hindi and English using existing models, or by translating English web data into Indian

languages to supplement low-resource languages. This synthetic data can address scarcity for smaller languages.

- **Data Cleaning & Augmentation:** Employ robust cleaning (e.g. Bloom filter for duplicate lines, profanity filters, normalization of scripts). Balance datasets: avoid over-representing English or popular languages by downsampling big corpora or up-weighting small ones. Augment data via paraphrasing (e.g. use an LLM to rephrase sentences) or **noising** (inject spelling variants common in Indic scripts). The *BhashaKritika* study suggests generating synthetic Indian text using LLMs to create ~540B tokens ³¹, a strategy India can emulate.
- **Domain Data:** For targeted applications, gather specialized data: medical texts in Hindi/English, legal proceedings, historical archives. This makes models useful in key sectors (healthcare, judiciary, education).

The strategy is to build a massive, balanced Indic corpus (text + code + maybe minimal common-sense knowledge bases) under an open license. India must aim for at least hundreds of billions of tokens (across languages) before training large models. Given current gaps ⁹, assembling new data (and synthetic generation) is as important as the modeling itself.

Compute Resources & Cost Planning

Training foundation models is compute-intensive. An individual engineer must carefully plan hardware and budgets:

- **GPU/Hardware Options:** Top choices are NVIDIA A100 or H100 GPUs (for double/triple precision and large memory). Typical current prices (mid-2026) are ~\$10–30K per GPU (e.g. H100 80GB ~\$25–35K ³²). Cloud alternatives (AWS, Azure, GCP) rent H100 instances for ~\$7–11 per GPU-hour ³³. Other options: NVIDIA L40S (48GB, ~\$6K), Tesla V100/A40 if cheaper but slower; TPUs (if Google partnership). On-prem solutions (DGX station, local clusters) require high upfront costs but lower per-hour once deployed.
- **Cloud vs On-Premises:** Cloud gives instant scalability: one can spin up dozens of A100/H100 GPUs. For example, Llama-2 70B used 2048 A100 GPUs for 23 days (~\$1.7M) ¹⁴. If hardware investment is impractical, use cloud batches (spot instances, reserved capacity). A hybrid is ideal: leverage IndiaAI's subsidized cloud (Jio, Tata, Yotta partnerships ³⁴) or educational clusters for initial prototyping, then scale on rented H100s for major runs. On-prem can be cost-effective if sustained large usage; e.g., 10 H100s (\$300K) at 24/7 usage for a year ~ \$2.6M, versus ~ \$4–6M equivalent cloud bill.
- **Compute Estimates:** Compute needs grow with model size and training tokens. Rough rule: training FLOPs $\approx 6 \times \text{\#params} \times \text{\#tokens}$. For example, training a 7B-parameter model on ~1T tokens might need $\sim 4 \times 10^{24}$ FLOPs – on the order of \$100K–\$500K with spot H100 pricing ¹⁴. In contrast, a 50B-parameter model on 5T tokens could be \$1–5M. Fine-tuning (with LoRA adapters) costs only ~1–5% of full training ³⁵. As a rule of thumb: **Low-budget (~₹5–10 lakh ≈ \$6K–\$12K)** can fine-tune a small open model (1–2B). **Mid-budget (~₹50–100 lakh ≈ \$60K–\$120K)** can train or fine-tune a 7–10B model on modest data. **High-budget (~₹5–10 crore ≈ \$0.6–1.2M)** can pretrain a ~30–50B model or many fine-tuning tasks. **Very-high (~₹50+ crore ≈ \$6M+)** would be needed for a 100+B training run from scratch. (For comparison, GPT-3 training cost ~\$4.6M, GPT-4 ~\$100M ⁸.)

- **Distributed Training:** For large runs, use data/model parallelism. Tools: PyTorch DDP, DeepSpeed (ZeRO), Megatron-LM, or JAX/pjit. Ensure high-speed networking (NVLink, InfiniBand). If on cloud, use clustered nodes with >100 Gbps interconnect. Efficient implementation (mixed precision, gradient checkpointing) can cut cost.
- **Cost Mitigation:** - Fine-tune pre-trained open models (e.g. Llama 3, Mistral) rather than full training when possible. Training from scratch should be reserved for models with unique Indian data. - Use academic/GPU grants (IndiaAI's 40% subsidy ³⁴, call for proposals), seek cloud credits (Google for Startups etc), or partner with companies that have unused GPU (e.g. gaming center rest times). - Incremental development: start with small models or modules (e.g. chat UI) to demonstrate value and secure more funding for bigger runs.

Team Structure and Talent Development

A successful LLM project requires a multidisciplinary team. Roles include:

- **ML Researchers/Engineers:** Experts in NLP and deep learning to design models, training regimes, and conduct experiments. They should handle architecture choices, optimization, and novel alignment techniques.
- **Data Engineers/Scientists:** To build and maintain the datasets pipeline: scraping, cleaning, augmenting multilingual corpora, and evaluating quality. They also curate benchmarks and annotation.
- **DevOps/MLOps:** Engineers to manage compute infrastructure (clusters, cloud), continuous integration of model training, and deployment pipelines for serving models and monitoring performance.
- **Domain Experts:** Specialists for Indian-specific content (linguists, ethicists, policy experts). They guide bias testing (e.g. handling caste questions), cultural relevance, and multilingual issues.
- **Safety/Alignment Team:** Focused on evaluation of model outputs for toxicity, bias, factuality, privacy leaks, and on implementing guardrails (filtering, post-processing, monitoring). This could include ethicists or social scientists.

Building this team involves: recruiting AI researchers (from universities, or returning diaspora), upskilling existing engineers (through IndiaAI's fellowship programs ²⁷, workshops, online courses). Partnerships with academia (IITs, IISc, IIITs) can provide interns or co-supervised PhDs. Encourage open collaboration by participating in global AI events and submitting research (publishing is key to attract talent).

To expand community talent: run workshops/hackathons on LLMs in major tech hubs (Bengaluru, Hyderabad, Delhi) and Tier-2 cities. Work with educational institutions to introduce practical AI courses. For example, IndiaAI's "FutureSkills" pillar funds PhDs and labs in IITs and NITs ²⁷. Mentorship programs (linking industry mentors to student teams) can also grow skills.

Funding Models and Partnerships

To sustain LLM development, a mix of funding and business models is needed:

- **Government Grants & Programs:** Continue leveraging IndiaAI grants, DST projects, and other R&D funding (MeitY, DBT, etc.). For example, Sarvam received GPU grants and cost-subsidy under IndiaAI ³⁴. Government can form consortia (like BharatGen) pooling academic and industry resources under public funding.

- **Venture Capital:** Encourage private investment. Sarvam's \$41M raise ¹⁸ shows VC interest in AI. More startups (beyond Sarvam/Krutrim) should pitch to VC/strategic investors. Highlight "AI for Bharat" opportunities (regional language services, enterprise AI in India).
- **Industry Consortium:** Involve large Indian tech companies (Tata, Jio, Wipro, Infosys) by creating a shared AI consortium. They could contribute compute credits, data, or expertise in exchange for stakes or usage rights. For example, NVidia is reportedly talking with Indian startups ²⁸. Industry partners can license finished models or integrate them into products.
- **Public-Private Partnerships:** Collaborations like UIDAI-Sarvam (Aadhaar AI) ³⁶ show how government (DPI) can adopt LLM features, funding development. Similarly, sectors like telecom (for voice assistants), banking, or agriculture can co-fund models tailored to their domain.
- **Open-Source and Service Models:** If models are open-sourced (Apache/MIT), revenues can come from support, fine-tuning services, or hosting (like Red Hat model). Alternatively, proprietary APIs (SaaS) can generate income but risk vendor-lock-in (Critics note SaaS doesn't build sovereignty ³⁷). A balanced approach: open core models (for transparency) with paid enterprise add-ons (security, compliance, customization).
- **Grants & Academic Funding:** Engage international grants (e.g. World Bank AI for Development, EU collaborations) and Indian science agencies (DST, DBT) for project funding. Also utilize programs like the IndiaAI Startups Global program (to go international) ³⁸.

In sum, multiple revenue streams – government, VC, industry contracts, and possible academic grants – should be combined. Early on, rely on grants and subsidies; as models mature, pivot to service/API models for sustainability.

Policy, Ethics, and Regulation

Developing LLMs in India must navigate legal and ethical landscapes:

- **Data Privacy:** India's forthcoming Personal Data Protection Bill (revised DPDP) will govern user data collection. LLM projects must ensure training data complies (anonymization, user consent where needed). Models shouldn't memorize personal data. Tools like differential privacy can help. Also, public data (social media, web crawl) may contain copyrighted or sensitive content; India should clarify fair-use for ML to avoid infringement.
- **Content Moderation and Bias:** Given India's diversity, models must handle sensitive topics carefully. The BIS is already planning standards for LLM output, specifically on caste, religion, gender bias ²⁹. Teams should proactively test and mitigate these biases. Developing "constrained decoding" or fine-tuning with guidelines (as in Constitutional AI) may be necessary. Open-sourcing datasets and evaluation sets (with community review) can improve transparency.
- **AI Ethics & Governance:** India can build on NITI Aayog's AI ethics principles, emphasizing fairness and transparency. A possible *IndiaAI Oversight Board* could audit high-risk AI systems, similar to "AI Safety Institute" being set up under IndiaAI ³⁹. Responsible use includes watermarking AI content (for misinformation concerns) and ensuring model provenance.
- **Regulatory Compliance:** Sector-specific rules (finance, healthcare) must be incorporated. For instance, AI in medical diagnostics requires adherence to national health data norms. Export

controls: training LLMs with US-origin GPUs may be restricted under US export rules; planning may require alternate sourcing.

- **Language and Accessibility:** Align with “Digital Public Infrastructure” (DPI) principles: Indian LLMs should be integrated into public services (e.g. e-governance portals) to maximize social benefit. As Sarvam’s CTO noted, pairing LLMs with India Stack services (Aadhaar, DigiLocker) could multiply impact ⁴⁰.

By embedding ethics and regulation from day one, India can avoid pitfalls faced elsewhere. Transparent reporting (e.g. following OECD’s Hiroshima AI reporting framework) will build trust.

Implementation Plan: Phases, Milestones, Risks & Mitigation

Phase I (0–3 months): Foundation.

- **Milestones:** Assemble core team; secure compute access (apply to IndiaAI, cloud grants); collect/clean initial datasets (Indic Wikipedia, government texts, news); train a small multilingual model (e.g. fine-tune LLaMA-2-7B on Indian data) to validate pipeline; engage with community (open GitHub repo, write blogs, join AI conferences in India). Develop basic benchmarks for Indian tasks.
- **Deliverables:** Proof-of-concept mini-model demonstrating multilingual output; public data corpus; project GitHub; initial project plan.
- **Risks:** Lack of compute access or data (mitigate by using public cloud trials, collaborating with universities/labs for resources); team inexperience (mitigate via online courses, mentors).

Phase II (3–12 months): Prototype Development.

- **Milestones:** Build larger model (10–30B parameters) using gathered corpora; implement instruction tuning on Indian QA data; launch small-scale pilot (e.g. chatbot for an Indian language); publish early results (blog/tech report); formalize collaborations (MoUs with IIT lab, tie-up with an ISP or govt data provider).
- **Deliverables:** Intermediate LLM (multi-lingual, context ~10k tokens), evaluation report on common-sense and Indian-specific QA tasks, small demo or API; contributed code/data to open source.
- **Risks:** Model underperformance (mitigate by iterating data cleaning and architecture tweaks); funding shortfall (seek small grants, pitch to local incubators); scope creep (focus on MVP functionalities first).

Phase III (1–2 years): Scaling Up.

- **Milestones:** Scale to a production-level model (50–100B) or use mixture-of-experts to increase capacity cost-effectively; integrate vision/speech if planned; rigorous safety testing; begin commercialization (e.g. licensing to a government project or enterprise pilot); form partnerships (e.g. with telecoms for localized voice assistant data).
- **Deliverables:** Mature LLM API or platform serving users; published papers or benchmarks demonstrating state-of-art on multilingual tasks; formal partnerships (e.g. with an NGO or PSU using the model). A business plan for sustainability.
- **Risks:** Technical obstacles in scaling (mitigate via academic collaboration, scaling core, efficient 3D parallelism); regulatory issues (mitigate by early dialogue with regulators); competition (stay agile, leverage local relevance).

Phase IV (3–5 years): Institutionalization.

- **Milestones:** Establish Indian LLM as a recognized tool (e.g. used in educational apps, government interfaces); spin off/expand team (new hires, possibly form a startup or research center); continue iterative improvement (e.g. train LLM 2.0 with more data, add more languages); influence policy (contribute to standards bodies).
- **Deliverables:** A version-2 model with industry adoption; a roadmap for next-generation models; ecosystem support (datasets, evaluation suites, developer community). Metrics showing impact (user growth, benchmarks achieved).
- **Risks:** Market/tech shifts (e.g. sudden new paradigm like retrieval RAG models – mitigate by staying flexible); resource exhaustion (maintain diversified funding, alliances).

Risk Mitigation (cross-cutting): Address compute risk by multi-sourcing (cloud credits, IIT cluster). Address talent risk via partnerships (IIT/industry) and remote collaborations (open labs). Data risk by synthetic augmentation and iterative data collection. Policy risk by proactive compliance (e.g. consult privacy lawyers, adhere to BIS guidelines).

KPIs and Success Metrics

To measure progress:

- **Model Performance:** Accuracy on benchmarks (MMLU, IndicGLUE, etc.), human evaluation scores, performance improvements over time.
- **Language Coverage:** Number of Indian languages supported, quality across languages.
- **Data Achievements:** Volume of corpus assembled (tokens, languages), number of high-quality curated datasets.
- **Compute Utilization:** GPU-hours used, cost efficiency (tokens trained per dollar).
- **Community Engagement:** GitHub commits/stars, downloads of models/datasets, number of contributors.
- **Deployment & Adoption:** Number of pilot projects/customers using the model, API calls served, app downloads if applicable.
- **Publications & Visibility:** Research papers, talks at conferences (incl. local events), mentions in media (e.g. TechCrunch on Sarvam).
- **Funding Milestones:** Grants secured, investment raised, partnerships formed.

Regularly tracking these (quarterly/annually) ensures accountability. For instance, IndiaAI publishes compute usage stats and fellowship counts ⁴¹ ²⁷ which could be adapted for project KPIs.

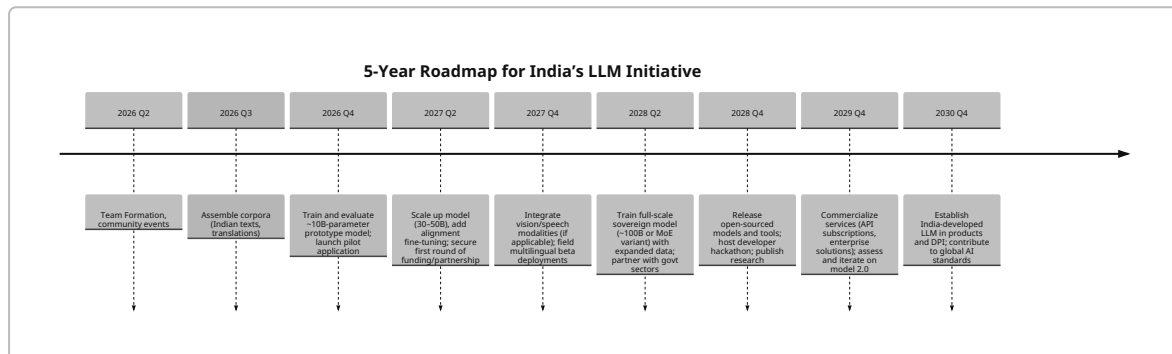
Partnerships & Ecosystem

Synergies are essential. Recommended partnerships include:

- **Industry:** Collaborate with Indian tech firms (Tata, Infosys, Wipro, Reliance, Ola/Via) for data-sharing and compute (e.g. Ola's GPUs). Cloud providers (AWS, GCP, Azure) for hosting/training credits. For language data, tie up with publishers (eg. newspapers, Encyclopaedia of India). Telecoms (Jio, Airtel) for multilingual customer service data. Banks/Insurance for financial text.
- **Academia:** IITs, IISc, TIFR, IIITs for research collaboration. Work with language institutes (C-DAC's TDIL program, Sahitya Akademi) for corpora. Engage universities in crowd-sourcing/annotation projects (perhaps as student projects).

- **Government:** Leverage bodies like NITI Aayog (policy), MeitY (funding), DST (research labs). Integrate models into Digital India platforms (Aadhaar, UPI) and citizen services. Contribute to standards groups (BIS, ISO) on AI norms.
- **International/Open:** Partner with open-source LLM communities (HuggingFace, BigScience) by contributing Indic data. Join international AI alliances (GPAI) as an Indian node. Use preprint servers (Arxiv India mirrors) to share progress.

These partnerships will pool resources and ensure the Indian LLM effort is well-integrated globally and locally.



First 90 Days and First Year Plan (Individual Engineer)

- **Days 1-30 (Orientation and Learning):** Set up a development environment (PyTorch, HuggingFace). Train a very small model (e.g. 100M parameter toy) on publicly available Indian text to learn the pipeline. Participate in online LLM forums (e.g. TensorFlow India, HuggingFace forums). Complete courses/tutorials on transformer training. Attend local AI meetups or webinars.
- **Days 30-90 (Data & Prototype):** Collect initial datasets (English Wikipedia + one Indian language). Clean and tokenize. Fine-tune an open model (like mBERT or Llama-7B) on this data. Build a simple demo (e.g. chatbot answering questions in Hindi). Document challenges. Reach out to mentors in AI4Bharat or local research groups. Apply for IndiaAI Compute access (or cloud credits). Submit a short technical blog about your prototype.
- **3-6 Months:** Expand datasets (more languages, sources). Re-train a larger model (7-10B parameters) using cloud GPUs. Measure performance vs baseline. Engage on GitHub (share code, data). Network: connect with Sarvam/Krutrim teams or similar startups; offer to help. Prepare a grant proposal (to university or startup incubator) for funding.
- **6-12 Months:** Secure necessary funding or compute (e.g. in an IndiaAI grant round). Collaborate to train a state-of-art model ($\geq 30B$) on specialized data. Publish results or present at a local conference. Launch an alpha product (chat interface or translation service) for user feedback. Recruit 1-2 volunteers or interns to help (maybe students). Begin discussions with potential industrial partners (e.g. a state govt interested in local language AI).

By the end of the first year, aim to have a functional multilingual model integrated into a real application (even if basic), a small team or collaboration network, and a clear roadmap with secured resources for

scaling up. Continuous learning and sharing (open blog posts, talks) will build credibility and attract collaborators.

Sources: Compilation based on industry reports and academic sources, including OpenAI release notes ¹⁰, Meta AI blog ³, government releases ⁴ ⁶, and research papers ⁹ ⁷.

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³ ³⁰ Introducing Meta Llama 3: The most capable openly available LLM to date

<https://ai.meta.com/blog/meta-llama-3/>

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⁵ IndiaAI Compute Capacity - OECD.AI

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⁷ Krutrim LLM: Multilingual Foundational Model for over a Billion People

<https://arxiv.org/html/2502.09642v2>

⁸ ¹⁴ ³⁵ How much does it cost to train an LLM on Cloud? GPT-4 training \$100M (as per Sam Altman) GPT-3 training \$5M LLaMa2 70b was trained on 2048 GPUs A100 for 23 days * \$1.5/hr per GPU = \$1.7M Cost to... | Lev Selector, Ph.D.

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¹⁰ Introducing GPT-4.5 | OpenAI

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¹¹ Ernie Bot - Wikipedia

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¹² Alibaba rolls out ChatGPT alternative Tongyi Qianwen · TechNode

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¹³ Over 30 AI models have been trained at the scale of GPT-4 | Epoch AI

<https://epoch.ai/data-insights/models-over-1e25-flop>

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²⁴ AI4Bharat

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²⁶ India's AI Talent Economy Enters A High Stakes Race For Skills And Salaries

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